ELECTROCHEMISTRY

Electrochemistry is the study of production of electricity from energy released due to spontaneous chemical reaction and the use of electrical energy to bring about spontaneous chemical transformations.

electric energy ⟷ chemical energy.

Electrochemistry

- electrochemical cell
  - Galvanic cell
    - ΔG = -ve
    - spon. chemical reaction
    - ↓
    - chemical energy → elec. energy
  - electrolytic cell (electrolysis)
- Conductance
  - ΔG = +ve
  - electric is given to current
  - non spontaneous rea^n.
  - electric energy
  - ↓
  - chemical energy

Electrochemical cells — these are the devices in which inter-conversion of electrical energy and chemical energy takes place.

Oxidation

- oxid^n is loss of e^-

$Zn^0 \rightarrow 2e^- + Zn^{2+}$

rxn in 0.5

Reduction

- red^n is gain of e^-

$Zn^{+2}_{(aq)} + 2e^- \rightarrow Zn^0_{(s)}$

- Red^n = tin oxidation no.

Oxidation state – is basically the count of electron that atom in a molecule can share, lose or gain while forming chemical bond.

$HClO_4$

$1 + x + (-2 \times 4) = 0$

$1 + x - 8 = 0$

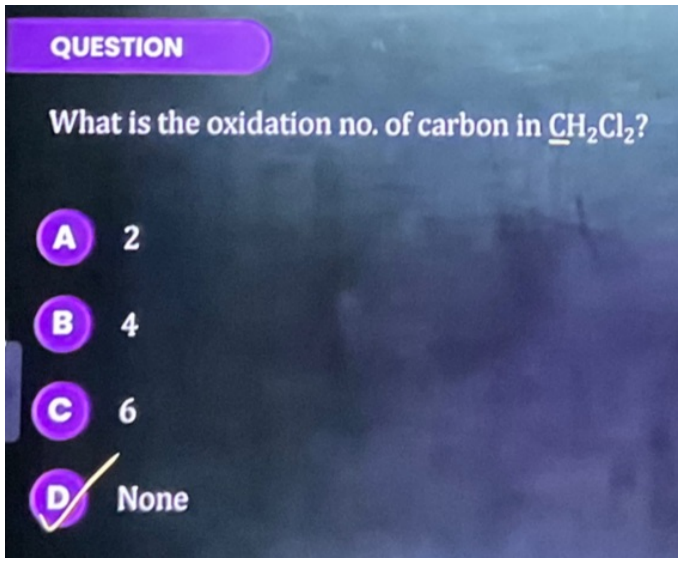
$x = +7$

$HClO_3$

$1 + x + (-2 \times 3) = 0$

$1 + x - 6 = 0$

$x = 5$.



Electrode – just a rod, which provide surface area for oxidation and reduction

- anode (A ↻ O) → always oxidn (loss of e^-)
- Cathode (C ↻ R) → always redn (gain of e^-)

Electrode

- active
  - participate in chemical rxn
  - Cu, Zn, Ag
- inert
  - don't participate
  - Pt, Au.

BM vidhan

1.) If not given, consider electrode to be inert

2.) If the given electrode is active, then all the cell reaction stop, and only, electrode will participate in chemical rxn.

Electrolyte – is a substance in which free ion are present, that's why it all allow electric current to pass through it (act as good conductor)

type of electrolyte

Strong
- mostly 100% ionisation
  - ↳ large no. of free ion
- eg: strong acid, base, salt, NaCl, HCl, H_2SO_4, NaOH

Weak
- Partial ionisation
  acetic acid, NH_4OH.

electrolyte

$NaCl \rightarrow Na^+ + Cl^-$
$MgCl_2 \rightarrow Mg^{+2} + 2Cl^-$
$HCl \rightarrow H^+ + Cl^-$
$H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}$

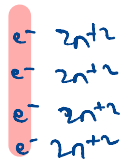
Non-electrolyte
↓
no-free ion
(act as bad conductor of electric current)
eg: urea, glucose

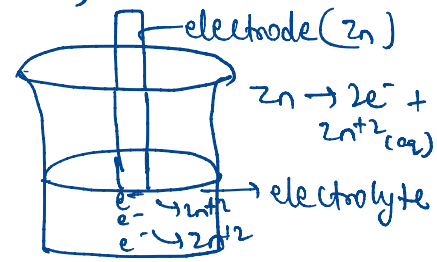
Redox reaction – reduction & oxidation

$Zn^0 + Cu^{+2} \longrightarrow Zn^{+2} + Cu^0_{(s)}$

decrease in O.S = redn
↑ in O.S = oxidn

Electrode potential – potential difference between electrode and electrolyte. (charge separation).





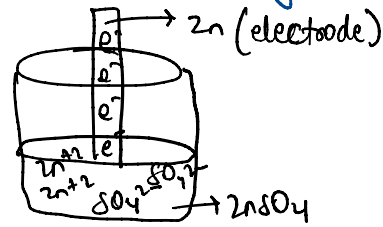
Case – I: $Zn_{(s)} \rightarrow 2e^- + Zn^{+2}$

(loss of e^-) (oxidn) → anode

solid Zn metal (electrodes) loses e^- and releases Zn^{+2} in aq. medium (electrolyte)

↓

toy to make electrode –ve charge

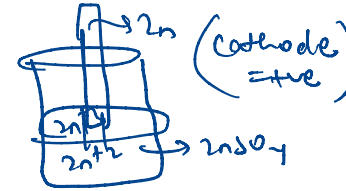


anode (oxid)
↳ –ve.

Case – II – cation from electrolyte move towards electrode and toy to make it +ve charge.

$Zn^{+2} + 2e^- \rightarrow Zn$

gain of e^- (red^n)



electrode potential – at each electrode – electrolyte interface there is a tendency of metal ions from the solution to deposit on the metal electrode toying to make it positively charged at the same time, metal atoms of the electrode have a tendancy to go into the sol^n as the ions

and leave behind the electrons at the electrode trying to make it negatively charged.

Standard Electrode Potential

There is separation of charges and depending on the tendencies of the two opposing reactions. The electrode maybe positively or negatively charged with respect to the solution. A potential difference develops between the electrode and the electrolyte which is called electrode potential when the concentration of all the species evolve in a half shell is unity. Then the electrode potential is known as the standard electrode potential.

Conc. = 1M
Temp = 25°C
Pressure = 1 bar

According to a IUPAC convention, standard reduction potential is are now called standard electrode potential in a galvanic cell and half cell in which oxidation takes place is called anode and it has a negative potential with respect to the solution. The other half cell in which the reduction takes place is called cathode and it has a positive potential with respect to the solution.

S.E.P = S.R.P.

Galvanic Cell

→ spontaneous reaⁿ
$\Delta G = -ve$

Electrolytic Cell

→ non-spontaneous reaⁿ
$\Delta G = +ve$

chemical energy ⟷ electrical energy

electric energy ⟷ chemical energy

LOAN — anode = −ve, cathode = +ve
left | oxidn | anode | −ve

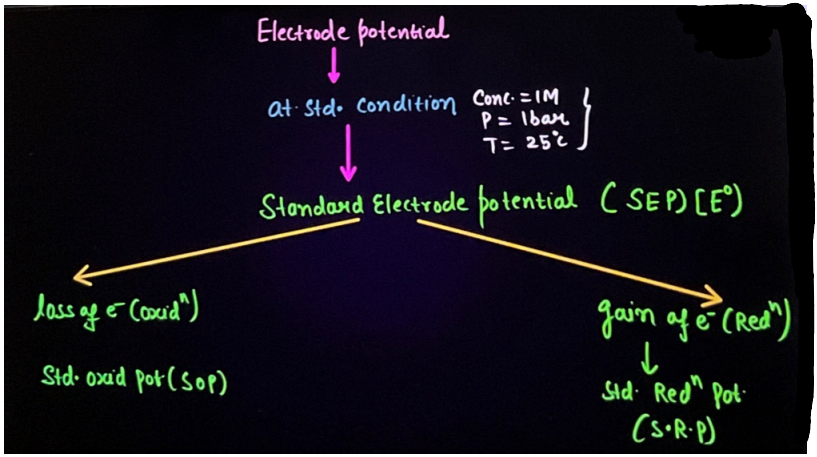
PANIC — cathode is +ve | anode | −ve

Electrode potential (E)

(power of electrode)

loss of e⁻ (oxidn is loss of e⁻) — oxidation pot.

gain of e⁻ (redn) — reduction pot.



1.) S.E.P ⟵ S.O.P / S.R.P } acc. to IUPAC

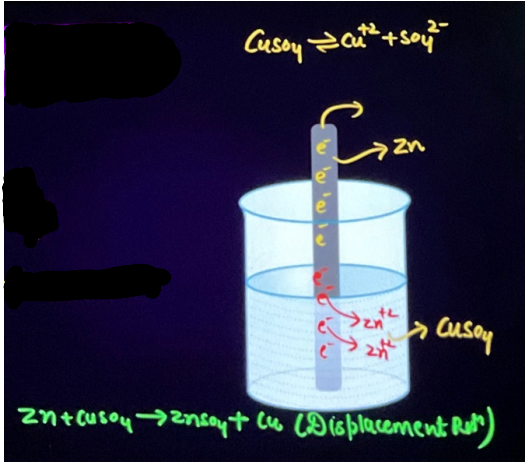
S.E.P = S.R.P

2.) S.R.P = −S.O.P.

Experiment - I

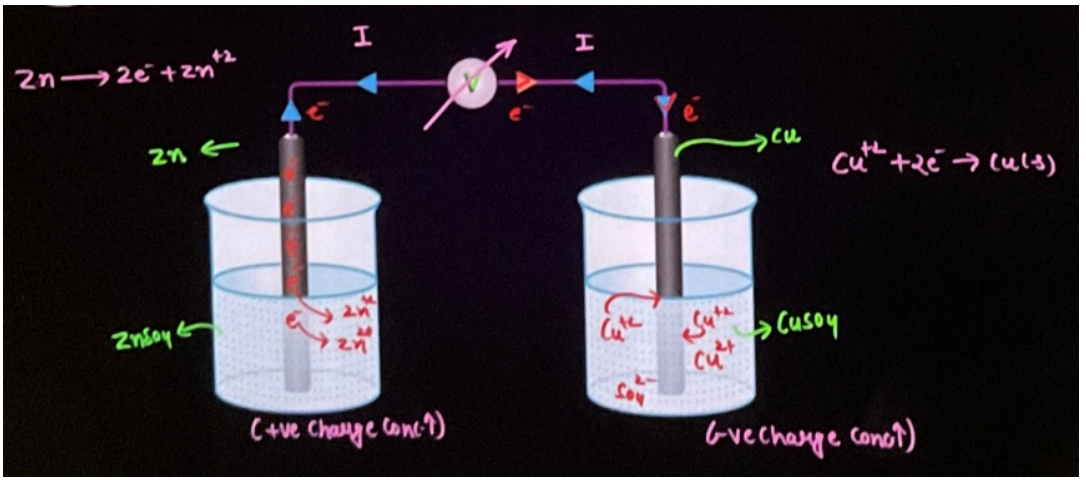
- spon. chemical reaⁿ
- chemical energy ⟷ electrical energy

condⁿ of electric current
* free ions / free e⁻



* Movement of e^-.

Experiment - II -



Experiment - III -

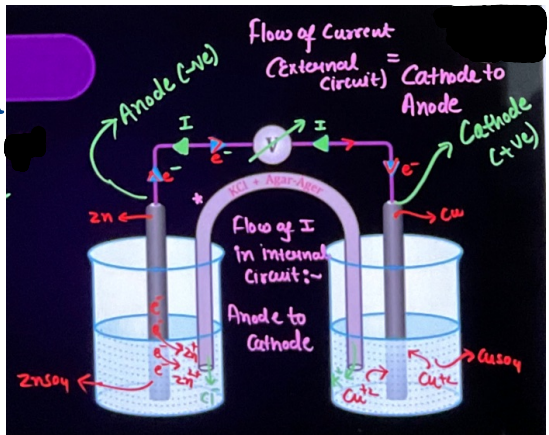
Salt bridge = KCl + agar-agar gel

Oxidn half cell : anode

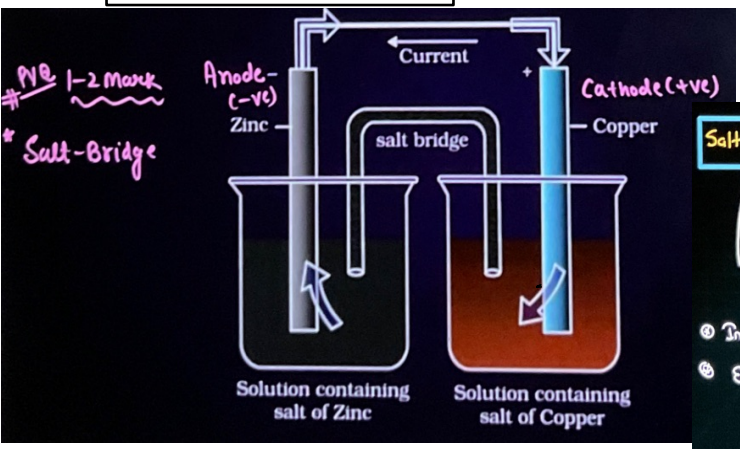
$Zn \rightarrow 2e^- + Zn^{2+}$ (ion of e^-)

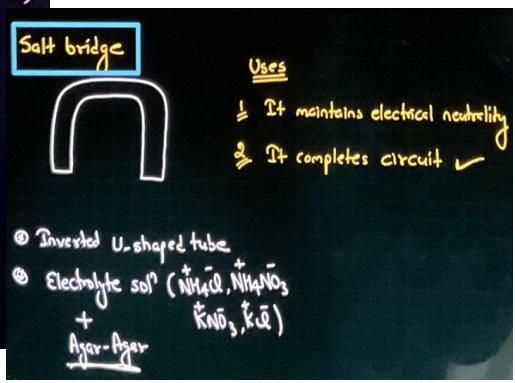
Redn half cell : Cathode

$Cu^{+2} + 2e^- \rightarrow Cu$.



Galvanic Cell





Cell reaction – 1.) anode half reaⁿ
2.) Cathode half reaⁿ
3.) Balance e^-
4.) Addⁿ of both half cell (for net cell reaction)

Net cell reaⁿ for galvanic cell.

1.) Anode half cell:

$Zn \rightarrow 2e^- + Zn^{+2}$

2.) Cathode half cell

$Cu^{+2} + 2e^- \rightarrow Cu$

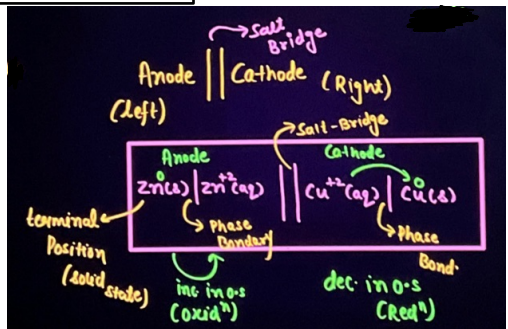
$$Zn \rightarrow 2e^- + Zn^{+2}$$
$$Cu^{+2} + 2e^- \rightarrow Cu$$
$$Zn^0 + Cu^{+2} \rightarrow Zn^{+2} + Cu^0$$

(net-cell reaⁿ)

redox reaⁿ.

Cell Representation.

galvanic cell
↓
L O A N
Left Anode



salt bridge
A B C
anode cathode

Cell Potential — potential diff. b/w both the electrode (anode & cathode)

(emf > pot. diff)

Cell Potential

- E.M.F (electromotive force) — max^{m} ability of cell to push e$^-$
  (when cell is not working)
  pot. diff. in open circuit
- Potential diff (real value)
  (working)
  Resistance, 'internal resistance'
  ↓
  Power drop.

10 hr → 2 hr. → red
max
Rel ✓

$$E_{cell} = E_{higher} - E_{lower}$$
$$E_{cell} = E_{cathode} - E_{anode}$$

Jiska redu^{n} pot. ↑ = red^{n} ↑ (cathode)

oxid. pot. ↑ = oxid^{n}
anode = oxid^{n} = oxid^{n}
↳ R.P (low) pot. ↑

Calculation of cell potential.

The cell potential is the difference between the electrode potential of the cathode and anode. It is called the cell electromotive force of the cell when no current is drawn through the cell.

$$E_{cell} = E_{right} - E_{left}$$

$$E_{cell} = E_{higher} - E_{lower}$$

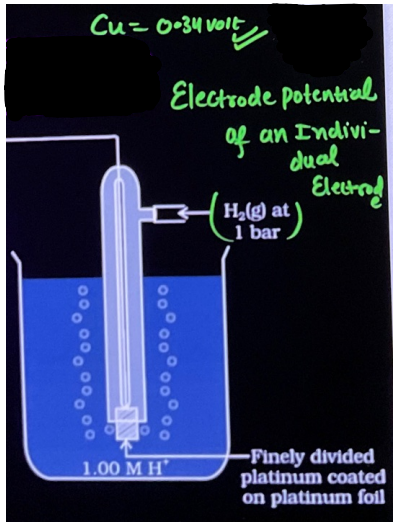
$$E^{\circ}_{cell} = E^{\circ}_{cathode} - E^{\circ}_{anode}$$

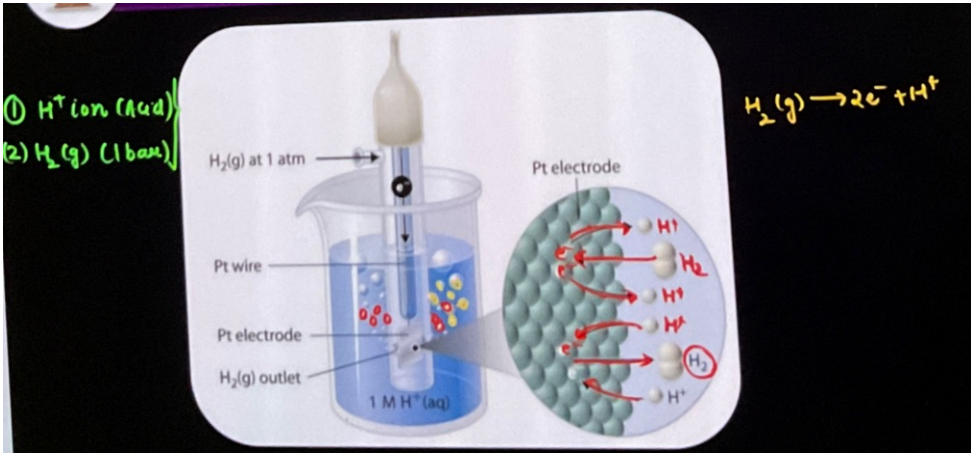
Measurement of electrode potential.

Standard hydrogen electrode

SRP of SHE has been assigned 0 V

The potential of individual half cell cannot be measured. We can measure only the difference between the 2 half cell potentials is that gives the EMF of the cell.





H^+ ion
H_2 gas

S.H.E ⟶ help electrode pot. ⇒ Std. electrode pot.

S.E.P of S.H.E = 0V

loss of e^- → S.H.E as anode

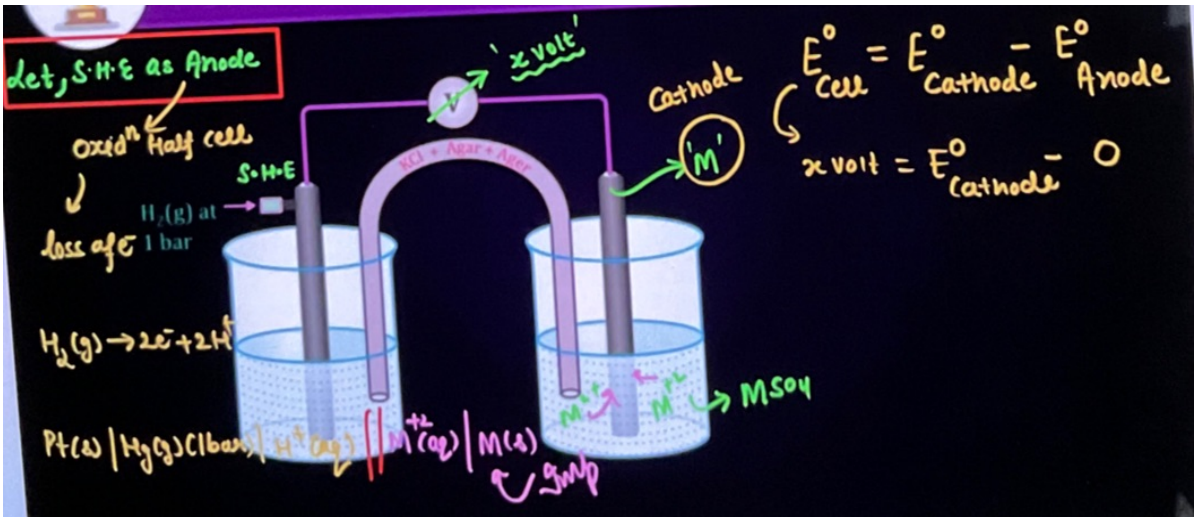
$H_2(g) \rightarrow 2e^- + 2H^+$

$Pt(s) \mid H_2(g)(1\,bar) \mid H^+(aq)$

gain of e^- → S.H.E as cathode

$2H^+ + 2e^- \rightarrow H_2\uparrow$

$Pt(s) \mid H^+(aq) \mid H_2(g)(1\,bar)$



Electrochemical cell and Gibbs Energy

ΔG = Gibbs free energy

ΔG° = std. gibbs free energy

ΔG° = -useful workdone by the sys.

In electrochemical cell
$\Rightarrow$ work done $\rightarrow$ elect. workdone

Pot. diff. $(\Delta V) = \frac{\text{workdone}}{\text{charge}}$

$$\Delta V = \frac{W}{q}$$

$$W = \Delta V \times q$$
$$W = E^\circ_{cell} \times q$$

q = charge

charge on 1 e^- = 1.6×10^{-19} C

charge on one mole e^- = $6.022 \times 10^{23} \times 1.6 \times 10^{-19}$

$1F \simeq 96500$

charge on 1 mole e^- = $1F = 96500$ C

" " n mole e^- = nF

$W = E_{cell} \times q$, $W = E \times nF$

$\Delta G^\circ = -W$ $\rightarrow$ $\boxed{\Delta G^\circ = -nFE^\circ_{cell}}$

Relation b/w common log and natural log.

Natural log = $\ln x$
↳ base = e

common log = $\log x$
↳ base = 10

$\log_e x = \ln x$

$\ln x = 2.303 \log x$.

Value of log

$e = 2.4$
$\log 1 = 0$
$\log 2 = 0.30$
$\log 3 = 0.47$
$\log 4 = 0.60$
$\log 5 = 0.69 \simeq 0.70$
$\log 10 = 1$

$\log_b b = 1$

Rule of log

1.) $\log m^n = n \log m$
2.) $\log (m.n) = \log m + \log n$
3.) $\log m/n = \log m . \log n$.

Nernst Equation

gas const. $(R) = 8.314\ J\ mol^{-1} K^{-1}$
temp $(x^\circ c)$

$$E_{cell} = E^\circ_{cell} - \frac{RT}{nF} \ln Q$$

E_{cell} ↓ at any conc.
E°_{cell} ↓ std. cond^n
n ↓ (e^-)
$F = 96500\ C$

E°_{cell} = at std. cond^n
conc. = unity (1M)

$\ln Q$ = natural log
Q = rx^n quotient.

$\ln Q = 2.303 \log Q$

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log Q$$

$$A + B \longrightarrow C + D$$

$$Q = \frac{[\text{Product}]}{[\text{Reactant}]} = \frac{[C][D]}{[A][B]}$$

Equilibrium constant from Nernst Equation

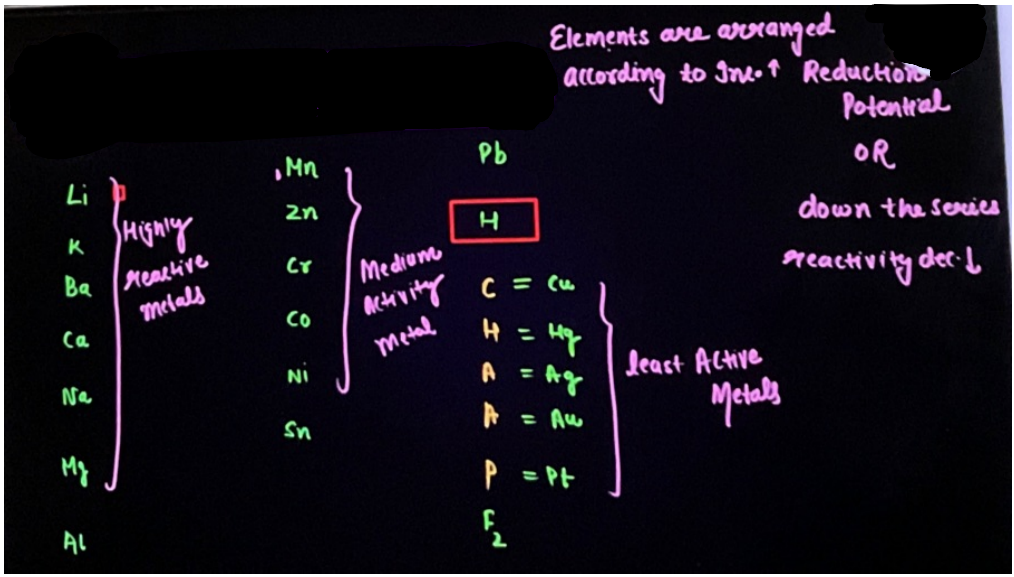
$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log Q$$

$$0 = E^\circ_{cell} - \frac{0.0591}{n} \log Q \quad (Q \to K_c)$$

$$E^\circ_{cell} = \frac{0.0591}{n} \log K_c$$

note: at equilibrium cond^n $E_{cell} = zero$

Electrochemical series



Reactive metals

red^n pot. = low

oxid^n pot. = high

self oxid^n — other = red^n

they are good reducing agent

} Reactive metals (R.P low)

H

} least active

in order of ↑ing red. pot.

R.P ↑ (high)

low activity metal
↓
red^n pot. = high
self = red^n
other = oxid^n
↓
oxidising agent

good reducing agent
reactive metals
or
(R.P low)

H

least active
they are good oxidising agent

in order of ↑ing red. pot.

R.P ↑ (high)

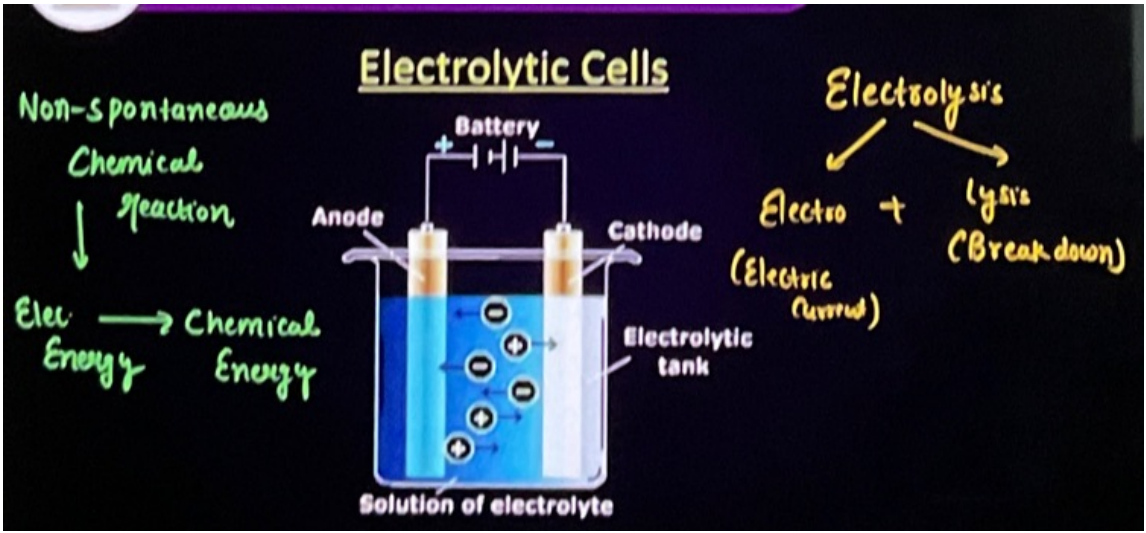
R.P = low
↳ O.P = high
↓
self = oxid^n
other = red^n

Note:—

R.P ↑ = reduction
O.P ↑ (high) = oxid^n
ΔG = gibbs free energy is additive in nature

$$\Delta G' = \Delta G_1 + \Delta G_2 + \Delta G_3 + \dots \Delta G_n$$

Electrolytic Cells and electrolysis

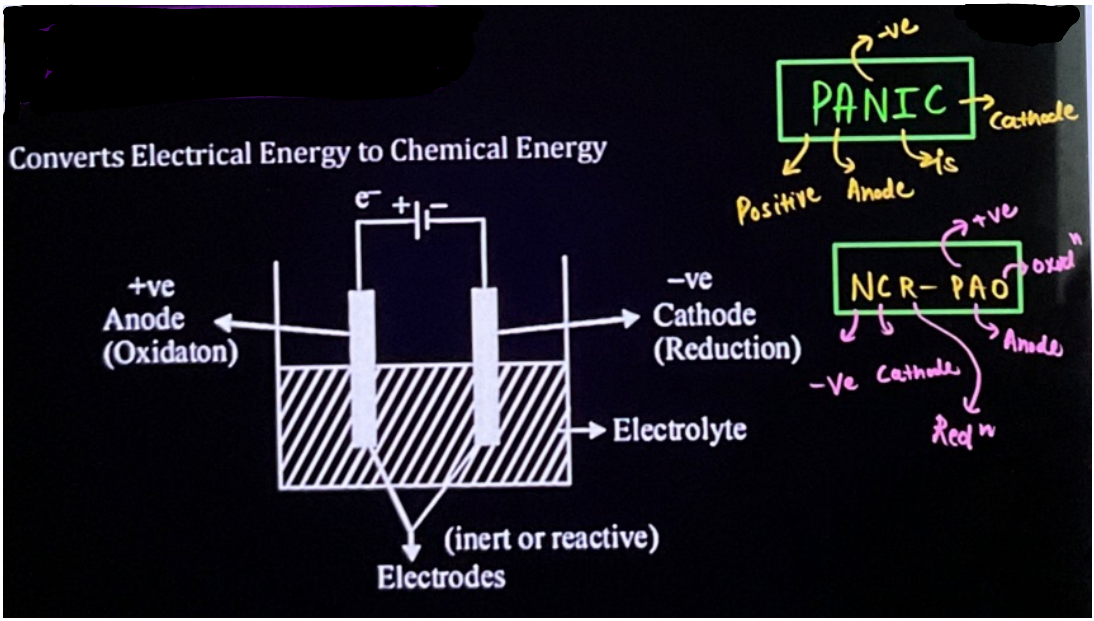


Electrolysis –

Decomposition of electrolyte solution by passage of electric current decomposition of metals or liberation of gases at electrodes takes place is known as electrolysis. In this process electrical energy is converted to chemical energy.

A nonspontaneous reduction action takes place in electrolysis with the help of battery or electrical energy.

Electrolysis cell



Products of Electrolysis

Products of electrolysis depend on the nature of material being electrolysed and the type of electrodes being used. If the electrode is inert, it does not participate in the chemical reaction. On the other hand. If the electrode is reactive it participate in the electrode reaction.

1.) Molten NaCl (no water present) $NaCl \rightleftharpoons Na^+ + Cl^-$

at anode : PAO (+ve) oxidⁿ (loss of e^-)

$2Cl^- \rightarrow 2e^- + Cl_2\uparrow$

at cathode : NCR (−ve) gain of e^-

$(Na^+ + 1e^- \rightarrow Na) \times 2$

$2Cl^- \rightarrow 2e^- + Cl_2$

$2Na^+ + 2e^- \rightarrow 2Na$

$2Na^+ + 2Cl^- \rightarrow 2Na + Cl_2\uparrow$

2.) aqueous NaCl → Cl^- will win.

at Anode (oxidⁿ) (Cl^-/OH^-)

$2Cl^- \rightarrow 2e^- + Cl_2\uparrow$

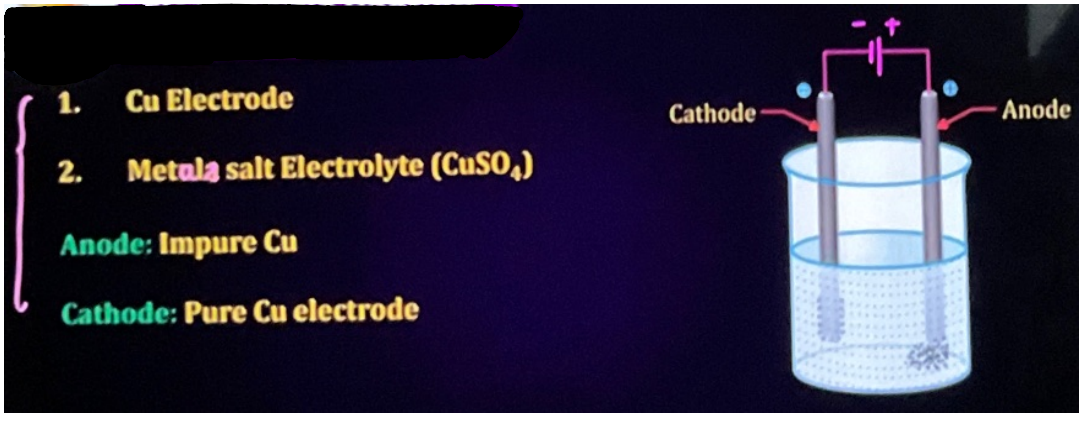
at cathode (redⁿ) (Na^+/H^+)

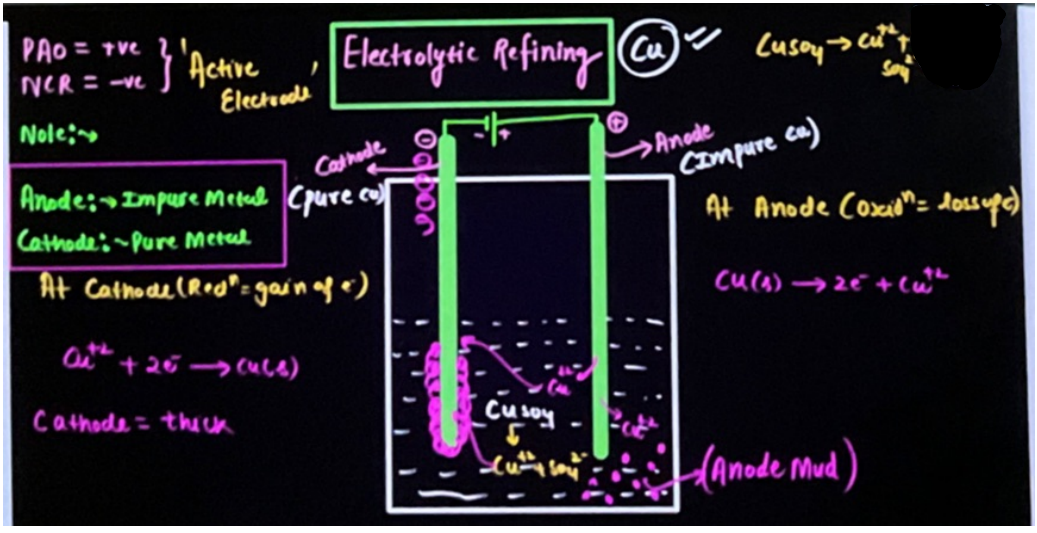
$2H^+ + 2e^- \rightarrow H_2$

$NaCl_{(aq)} \rightleftharpoons Na^+ + Cl^-$

$H_2O \rightleftharpoons H^+ + OH^-$

Electrolytic Reaction





Faraday's law of electrolysis

First law: the amount of chemical reaction which occurs at any electrode during electrolysis by a current is proportional to the quantity of electricity passed through the electrolyte

$$W \propto q$$

$$q = I \times t$$

$$W = Z \times I \times t$$

W: amount of product; Z: electrochemical eq.; I: current; t: time

$$Z = \frac{E}{96500}$$

E: equivalent wt.

$$\text{eq. wt.} = \frac{\text{Molar mass}}{\text{n-factor}}$$

- $HCl \rightarrow H^+ + Cl^-$, n-factor = 1
- acid (H^+ ion per acid molecule)
- Base (OH^-) per base molecule
- Salt = total +ve

Second law: the amount of different substances librated by the same quantity of electricity, passing through the electrolytic solution are proportional to their chemical equivalent weights

Case-I

$W_1 = Z \times I_1 \times t_1$

$Z = \frac{E_1}{96500}$

Case-II

$W_2 = Z \times I_2 \times t_2$

$Z = \frac{E_2}{96500}$

$\frac{W_1}{W_2} = \frac{E_1}{E_2}$

Type of Conductor

Differences between Metallic and Electrolytic Conduction.

	Metallic Conduction	Electrolytic Conduction
1.	Metallic conduction is carried by the movement of electrons.	Electrolytic conduction is carried by the movement of ions.
2.	Evolves no change in the chemical properties of the conductor.	It involves the decomposition of the electrolyte as a result of the chemical reaction.
3.	It does not involve the transfer of any matter.	It involves the transfer of matter as ions.
4.	Metallic conduction decreases with increase in temperature.	Electrolytic conduction increases with increase in temperature.

Relation b/w diff. terms.

$R = \rho \frac{L}{A}$

R = resistance

ρ = resistivity

l = length of conductor

A = area of cross-section.

$\frac{1}{\rho} = \frac{1}{R}\frac{l}{A}$

$K = G \times G^*$

Kappa

K = conductivity

G = conductance

G^* = cell constant

Conductance

$G = \frac{1}{R} = \frac{1}{\Omega} = \Omega^{-1}$ / S / mho

$R = \rho \frac{l}{A}$

$\frac{1}{R} = G$

G = Conductance

→ Rate of flow of charge in an Electrolytic soln

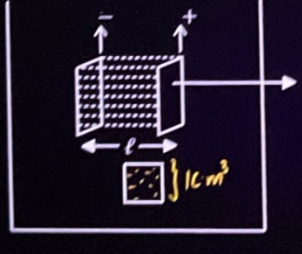


} Comp. Electrolyte (Rate of flow of charge)

Conductivity

$K = G \times G^*$

Rate of flow of charge in $1 cm^3$ Volm



Conductance of 1 cm^3 or 1 m^3 solution is called conductivity (K)

Conductance (G)

→ Rate of flow of Charge in Electrolytic soln

$G = \frac{1}{R} = \frac{1}{\Omega} = \Omega^{-1}$

Conductivity (K)

In → $1 cm^3$ Rate of flow of charge

$R = \rho \frac{l}{A}$

$\frac{1}{\rho} = \frac{1}{R} \frac{l}{A}$

$K = \Omega^{-1} \frac{cm}{cm^2}$

$K = \Omega^{-1} cm^{-1}$

$= S cm^{-1}$

Molar Conductance (Λ_m)

→ Rate of flow of Charge in 1 Mole Electrolyte having V cm^3 Volume

$\Lambda_m = \frac{K \times 1000}{M}$

$\Lambda_m = \frac{S cm^{-1}}{mol / cm^3}$

$S cm^2 mol^{-1}$

Equivalent Conductance (Λ_N)

$1 eq / \Lambda_N$

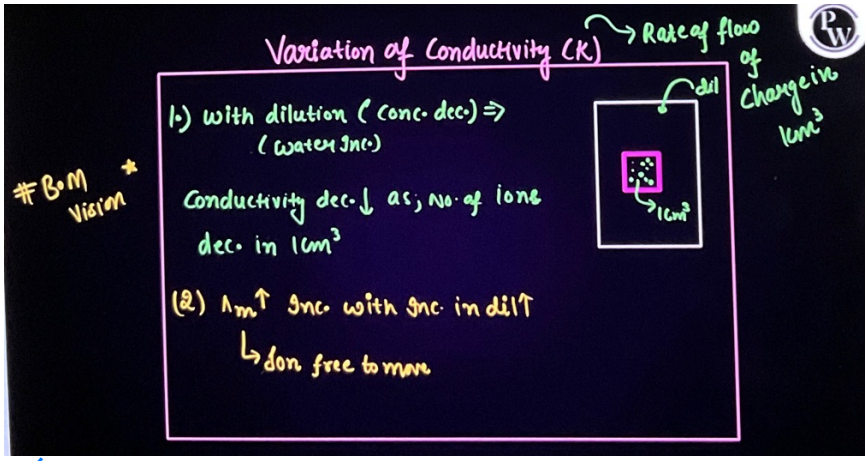
→ Rate of flow of charge in one normal Electrolyte

$\Lambda_N = \frac{K \times 1000}{N}$ ↳ soln

Kw 'Unit'

Variation of conductivity and molar conductivity with concentration.

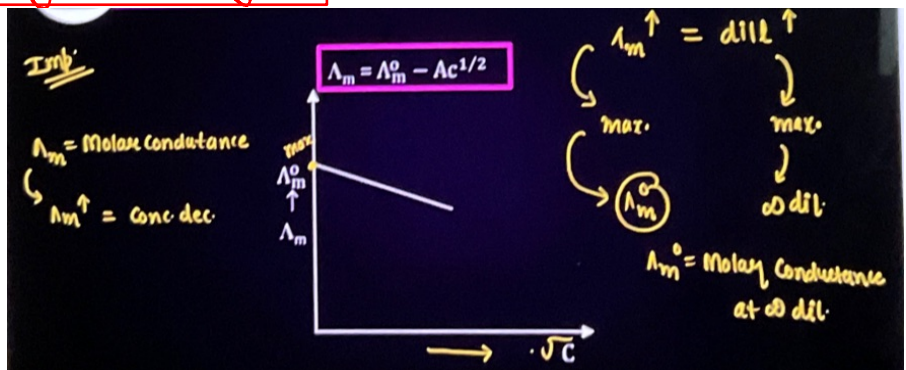
Conductivity Decreases with decrease in concentration for both type of electrolytes, weak or strong. Molar conductivity increases with decrease in concentration for both type of electrolyte



Explanation:

when we add solvent in solution number of ions per unit volume decreases so conductivity decreases but due to dilution distance between cation and anion increases so due to this attraction force between them will be less. So now I ions can move easily towards opposite nature of electrodes. So molar conductivity increases.

Strong electrolytes



NCERT

Strong Electrolytes (Conc. dec.)

For strong electrolytes, Λ_m increases slowly with dilution and can be represented by the equation:

$$\Lambda_m = \Lambda_m^\circ - A\, c^{1/2} \quad (2.23)$$

It can be seen that if we plot (Fig. 2.6) Λ_m against $c^{1/2}$, we obtain a straight line with intercept equal to Λ_m° and slope equal to '$-A$'. The value of the constant 'A' for a given solvent and temperature depends on the type of electrolyte i.e., the charges on the cation and anion produced on the dissociation of the electrolyte in the solution. Thus, NaCl, $CaCl_2$, $MgSO_4$ are known as 1-1, 2-1 and 2-2 electrolytes respectively. All electrolytes of a particular type have the same value for 'A'.

PYQ

PYQ

Imp

NCERT

↑dil. α degree of diss ↑

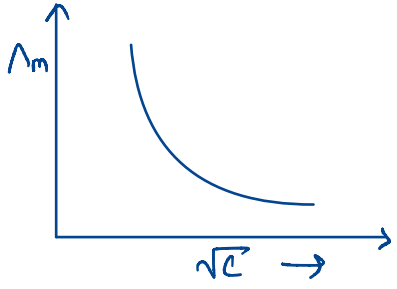
Weak Electrolytes

Weak electrolytes like acetic acid have lower degree of dissociation at higher concentrations and hence for such electrolytes, the change in Λ_m with dilution is due to increase in the degree of dissociation and consequently the number of ions in total volume of solution that contains 1 mol of electrolyte. In such cases Λ_m increases steeply (Fig. 2.6) on dilution, especially near lower concentrations. Therefore, Λ_m cannot be obtained by extrapolation of Λ_m to zero concentration. At infinite dilution (i.e., concentration $c \to$ zero) electrolyte dissociates completely ($\alpha = 1$), but at such low concentration the conductivity of the solution is so low that it cannot be measured accurately. Therefore, Λ_m° for weak electrolytes is obtained by using Kohlrausch law of independent migration of ions (Example 2.8). At any concentration c, if α is the degree of dissociation

PYQ

A/R

Weak electrolytes — α = more ↑



Kohlrausch law of independent Migration of ions

At infinite dilution, ions, move freely that means movement of ions does not depend upon the presence of co-ions. Molar conductivity of an electrolyte at infinite dilution, is sum of molar conductivities of its constituent ions in the electrolyte

$$\Lambda^\circ_{m(NaCl)} = \Lambda^\circ_{m(Na^+)} + \Lambda^\circ_{m(Cl^-)}$$

at infinite dil^n, each ion has its own characteristic conductance
at infinite dil^n, weak electrolyte is 100% dissociated.

Primary batteries

In the primary batteries, the reaction occurs only once and after use over a period of time, battery becomes dead and cannot be reused again. The most familiar example of this type is the dry cell known as the Leclanché cell which is used commonly in our transistors and clocks.

Construction of cell

In the primary batteries, the reaction occurs only once and after use over a period of time battery becomes dead and cannot be reused again.
The most familiar example of this type is the dry cell (known as Leclanche cell after its discoverer) which is used commonly in our transistors and clocks.

Anode : $Zn(s) \rightarrow Zn^{2+} + 2e^-$ (Oxidn, O.S ↑)

Cathode : $MnO_2 + NH_4^+ + e^- \rightarrow MnO(OH) + NH_3$ (+4 → +3)

(Redn) dec in O.S = Redn



Mercury Cell

Anode ✓
Cathode ✓

Imp

Mercury cell, suitable for **low current devices** like hearing aids, watches, etc. consists of zinc - mercury amalgam as anode and a paste of HgO and carbon as the cathode. The electrolyte is a paste of KOH and ZnO

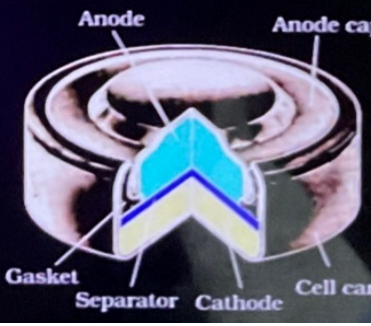
oxid. (inc. in o.s)

Anode : $Zn(Hg) + 2OH \rightarrow ZnO(S) + H_2O + 2e^-$

Cathode : $HgO + H_2O + 2e^- \rightarrow Hg(l) + 2OH^-$

(Redⁿ)

dec in o.s = Redⁿ



lead storage battery

commonly used in automobiles and invertors. It consists of a lead anode and a grid of lead packed with lead dioxide (PbO_2) as cathode. A 38% solution of sulphuric acid is used as an electrolyte.

electrolyte

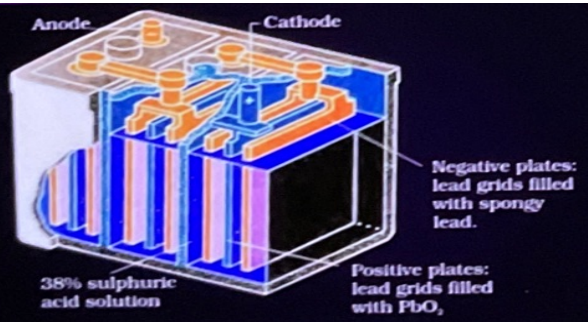
oxidⁿ

Anode : $Pb(S) + SO_4^{2-}(aq) \rightarrow PbSO_4(S) + 2e^-$

Cathode : $PbO_2(S) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightarrow PbSO_4(S) + 2H_2O(l)$

dec. in o.s

discharging cell reactions

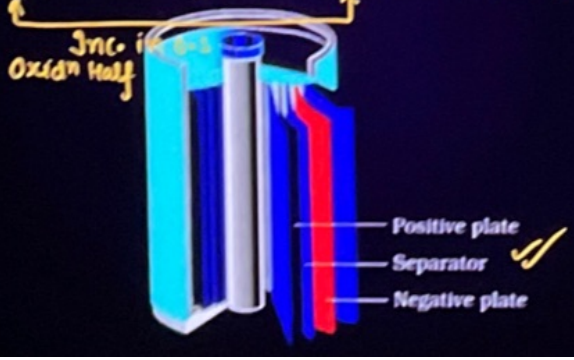


On charging the battery the reaction is reversed and $PbSO_4(s)$ on anode and cathode is converted into Pb and PbO_2, respectively.

Nickel – Cadmium Cell

Nickel-cadmium cell has longer life than the lead storage cell but more expensive to manufacture

$$Cd(s) + 2Ni(OH)_3(s) \longrightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O(l)$$



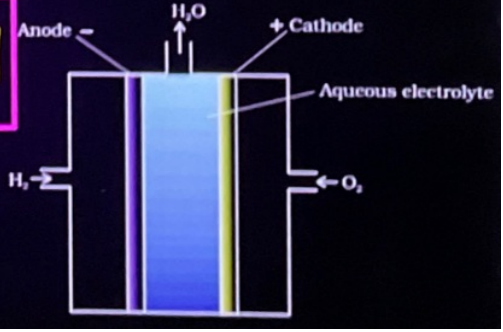
Fuel Cells

Galvanic cells that are designed to convert the energy of combustion of fuels like hydrogen, methane, methanol, etc. directly into electrical energy are called fuel cells.

Energy of combustion → Electrical Energy

Cathode : $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$

Anode : $2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(l) + 4e^-$



THANK YOU !!